

Introduction to Artificial Intelligence

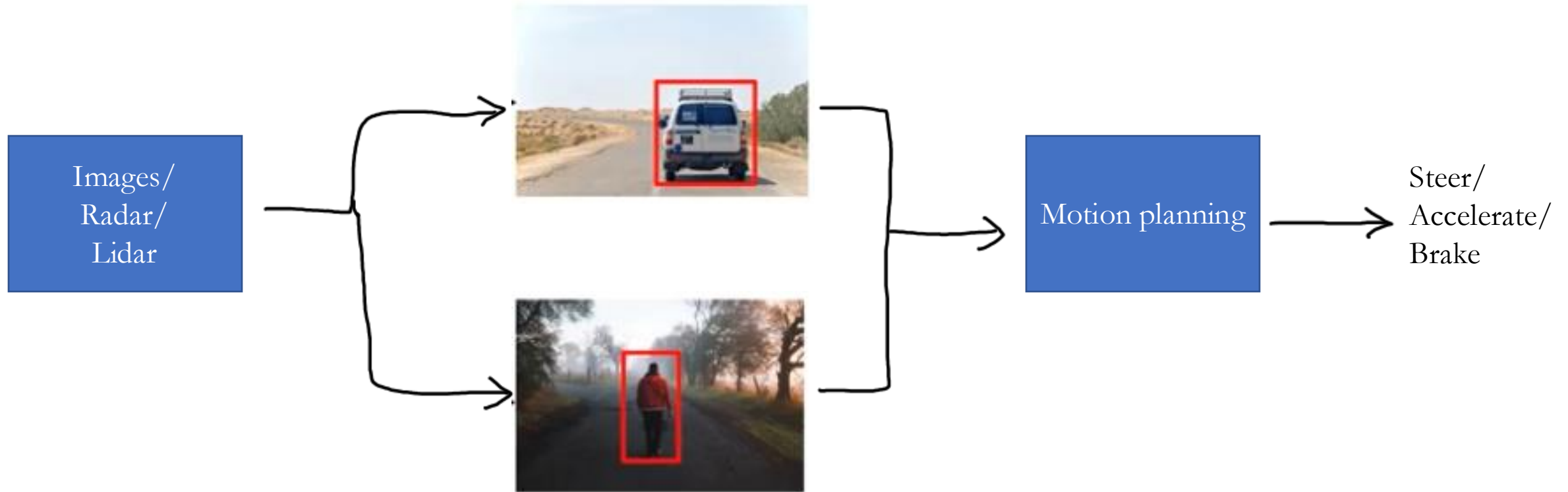
Lecture 13: Intelligent search II

October 20, 2025



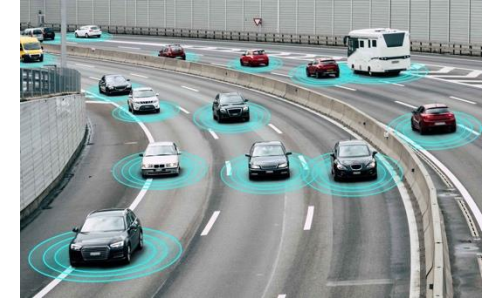
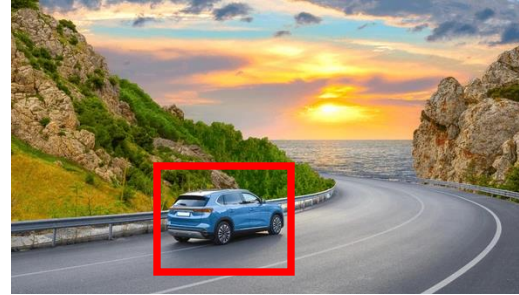
Case study: self-driving cars

- Steps for deciding how to derive

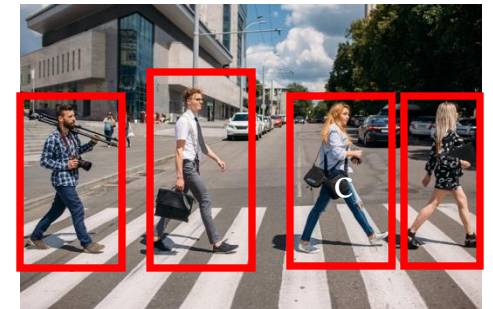


Key steps

1. Car detection



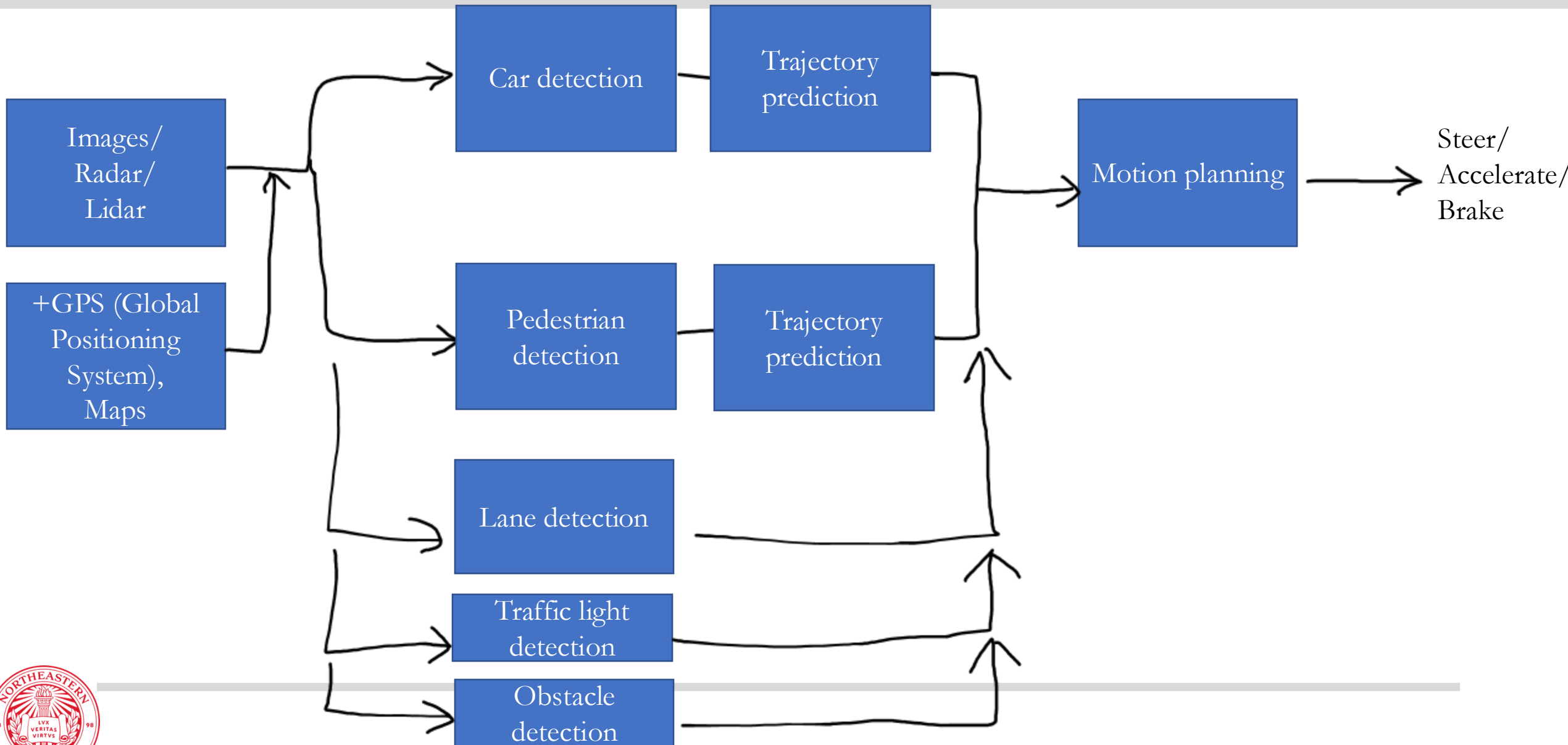
2. Pedestrian detection



3. Motion planning: tell you what is the path to move forward



Steps for deciding how to drive



Major AI application areas: Computer Vision

- Image classification/Object recognition
 - Face recognition

register



new



Cat

- Object detection



- Image segmentation



- Tracking



Natural Language Processing

- Text classification
 - Sentiment recognition
- Information retrieval
 - E.g., web search
- Named entity recognition
- Machine translation

Email → Spam/Non-Spam
Product description → Product category
“The food was good” → 4/5
“Service was horrible” → 1/5

Washington — Sen. *Mark Kelly*, an
Arizona Democrat, said Sunday that
lawmakers could resolve the shutdown
stalemate and reopen the government as
soon as this week, urging Republicans to
come to the negotiating table on health
care subsidies.

English - Detected Chinese (Simplified) English Korean

Spanish Chinese (Simplified) English

AI is the new electricity



人工智能是新电力



Speech

Variation in tone

How air pressure is
changing



time

- Speech recognition (speech-to-text)
- Trigger word/wake-word detection: alexa, hey google, hi device
- Speaker ID: listen to someone speak and verify the identity of speaker
- Speech synthesis (text-to-speech, TTS)
 - The quick brown fox jumps over the lazy dog.

Generative AI

- Artificial intelligence systems that can produce high quality content, specifically **text**, **images**, and **audio**

- Text generation: “What is AI?” —————>

Gemini: a broad field of computer science focused on creating systems that can perform tasks that typically require human intelligence. This includes abilities like learning from experience, reasoning, solving problems, understanding language, and perceiving the environment.

- Image generation: “a purple friendly robot eating ice-cream” —————>



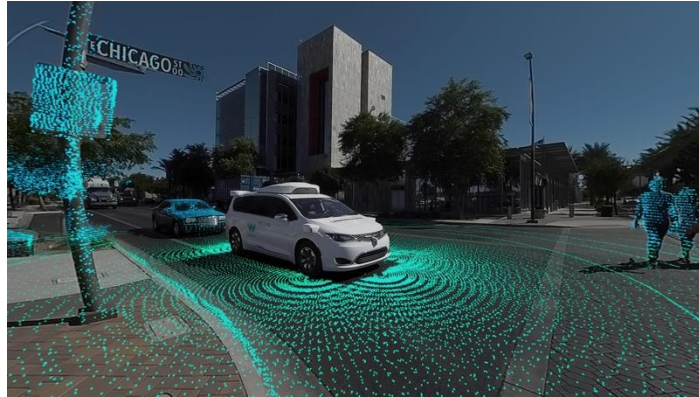
- Audio generation

- Speech, music: “You Are Not Alone” —————>



Robotics

- Perception: figuring out what's in the world around you



- Motion planning: finding a path for the robot to follow
- Control: sending commands to the motors to follow a path

General machine learning

- Unstructured data (images, audio, text)



Washington — Sen. Mark Kelly, an Arizona Democrat, said Sunday that lawmakers could resolve the shutdown stalemate and reopen the government as soon as this week, urging Republicans to come to the negotiating table on health care subsidies.

- Structured data

Size of house (square feet)	# of bedrooms	Price (1000\$)	User ID	Time	Price (1000\$)	Starred
523	1	305	1234	Aug 3	490	yes
645	1	384	9871	Sep 5	599	no
726	2	540	7623	Sep 20	384	yes
1088	2	653	1097	Oct 2	999	no



Lecture plan

- Intelligent search
 - Wrapping up un-informed search
 - Informed search algorithms: greedy, A^* search



Example search problems

- Traveling salesperson problem (TSP): find a minimum cost traveling tour across all the locations
- VLSI layout problem: positioning millions of components and connections on a chip to minimize area, minimize circuit delays, minimize stray capacitances, and maximize manufacturing yield
 - Real robots must deal with errors in their sensor readings and motor controls, with partial observability, and with other agents that might alter the environment
- Automatic assembly sequencing: find an order in which to assemble the parts of some object
 - If the order is wrong, there will be no way to add some part later in the sequence without undoing some of the work already done



Summary of uninformed search algorithms

Scenario	Best Algorithm	Why?
Equal step costs, shallow solution	Breadth-first search	Guarantees shortest path by steps
Memory limited, deep tree	Depth-first search	Linear space complexity
Varying costs, need optimal solution	Uniform cost search	Finds least-cost path
Need all shortest paths	Dijkstra's algorithm	Computes complete shortest path tree



Informed search algorithms

- With the aid of domain-specific knowledge
 - Can help people solve hard problems without search
 - Can help computers find solutions more efficiently
- Informed heuristic search
 - Estimate the cost from a given node to a goal node
 - Take the estimate towards the goal into account when selecting the path
 - Cost function: $f(n) = g(n) + h(n)$
 - $g(n)$: actual cost from start to node n
 - $h(n)$: heuristic cost from node n to the target



Cost function

The choice of f determines the search strategy:

- Uniform cost search: $f(n) = g(n)$ (expand lowest path cost)
- Greedy search: $f(n) = h(n)$ (expand closest to goal)
- A^* : $f(n) = g(n) + h(n)$ (balance both factors)
- Weighted A^* : $f(n) = g(n) + w \times h(n)$



Heuristic function

- Heuristic function: $h(n)$ = estimated cost of the cheapest path from the state at node n to a goal state
 - $h(n)$ is arbitrary, non-negative, and problem-specific
 - If n is a goal node, $h(n) = 0$
 - $h(n)$ must be easy to compute (without search)
- Common heuristics functions for path planning
 - **Manhattan Distance:** $h(n) = |x_1 - x_2| + |y_1 - y_2|$
 - **Euclidean Distance:** $h(n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
 - **Diagonal Distance:** $h(n) = \max(|x_1 - x_2|, |y_1 - y_2|)$



Example: Path planning in Romania



Abstracted graph representations

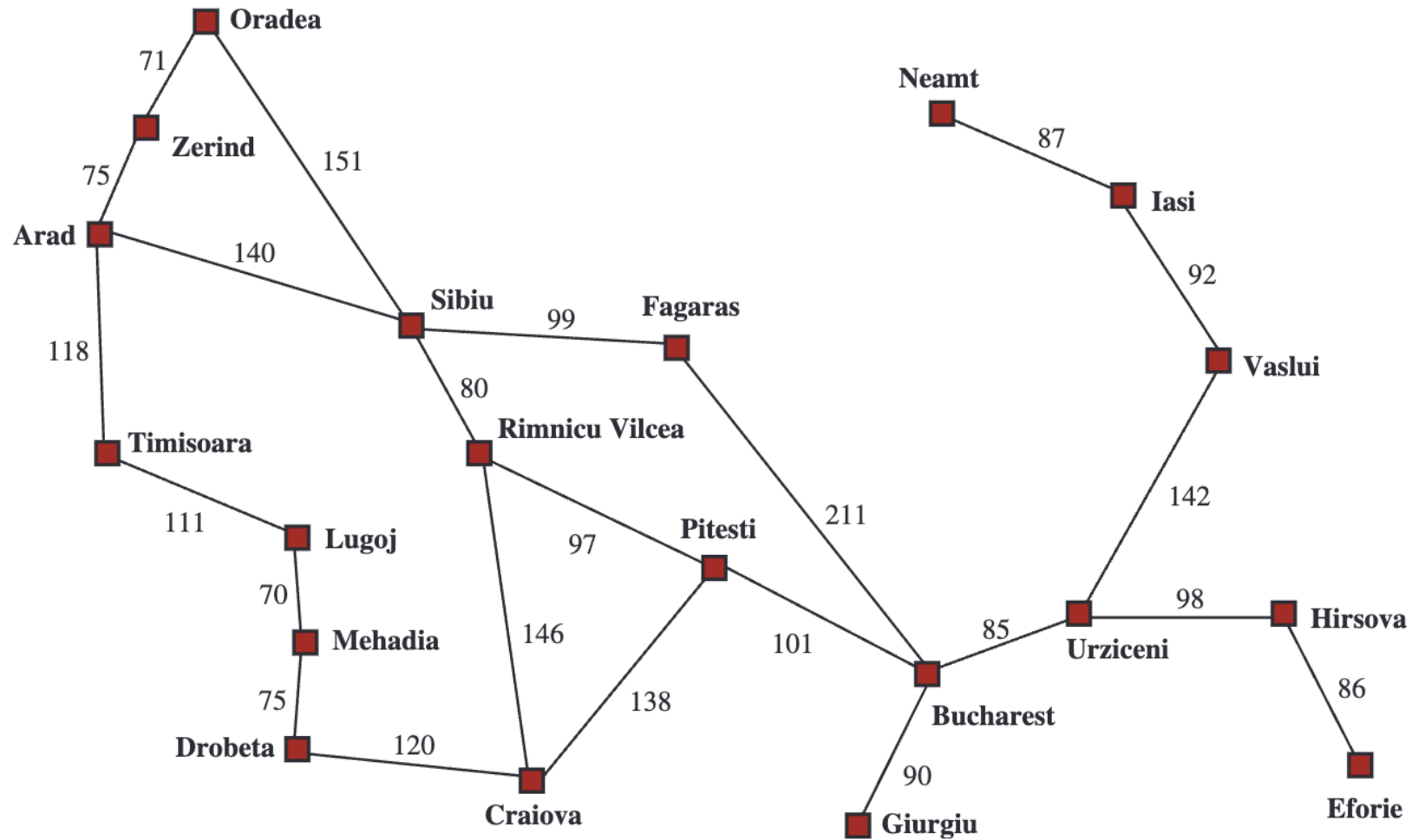


Illustration of greedy search

- Goal: minimize the estimated cost to the goal, $f(n) = h(n)$
- E.g. $h_{\text{SLD}}(n)$ = straight-line distance from n to Bucharest

Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
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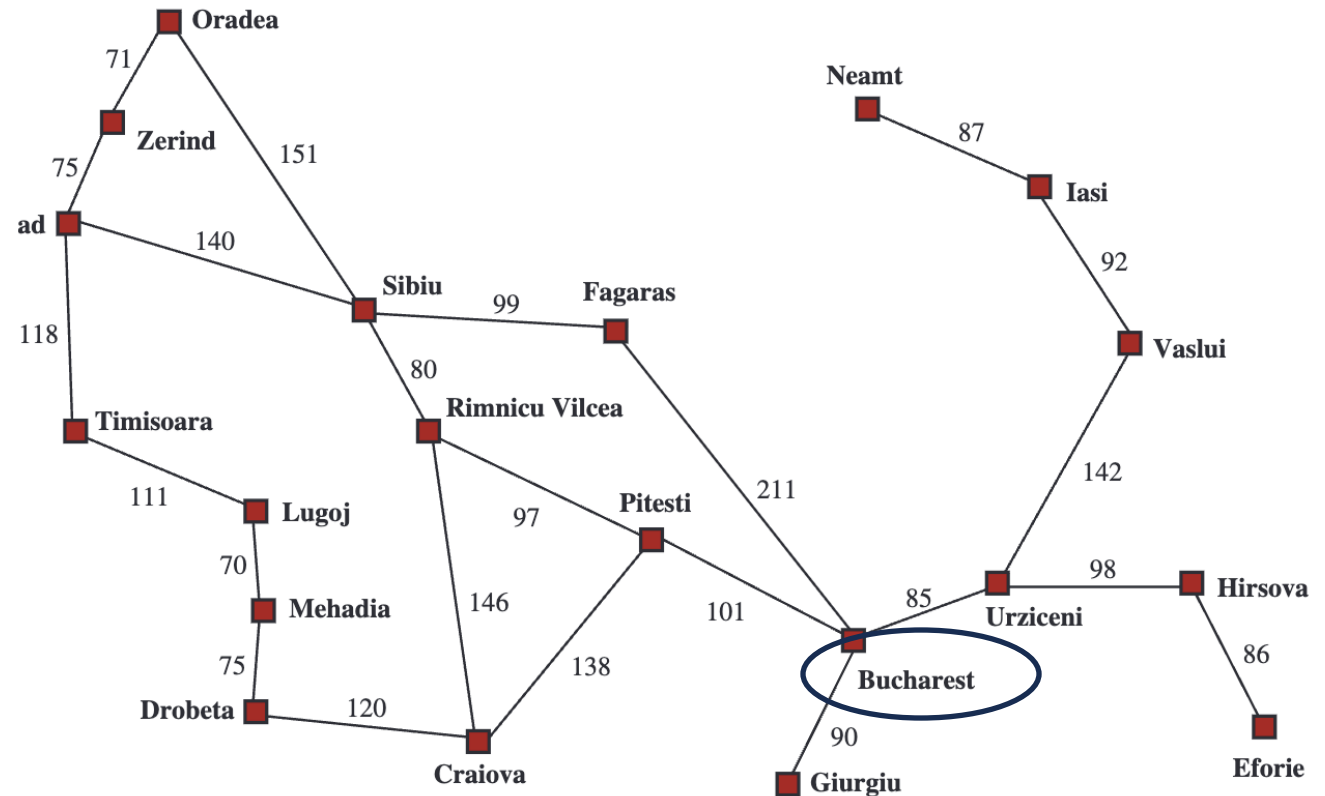


Illustration of greedy search

- Greedy search expands the node that appears to be closest to goal

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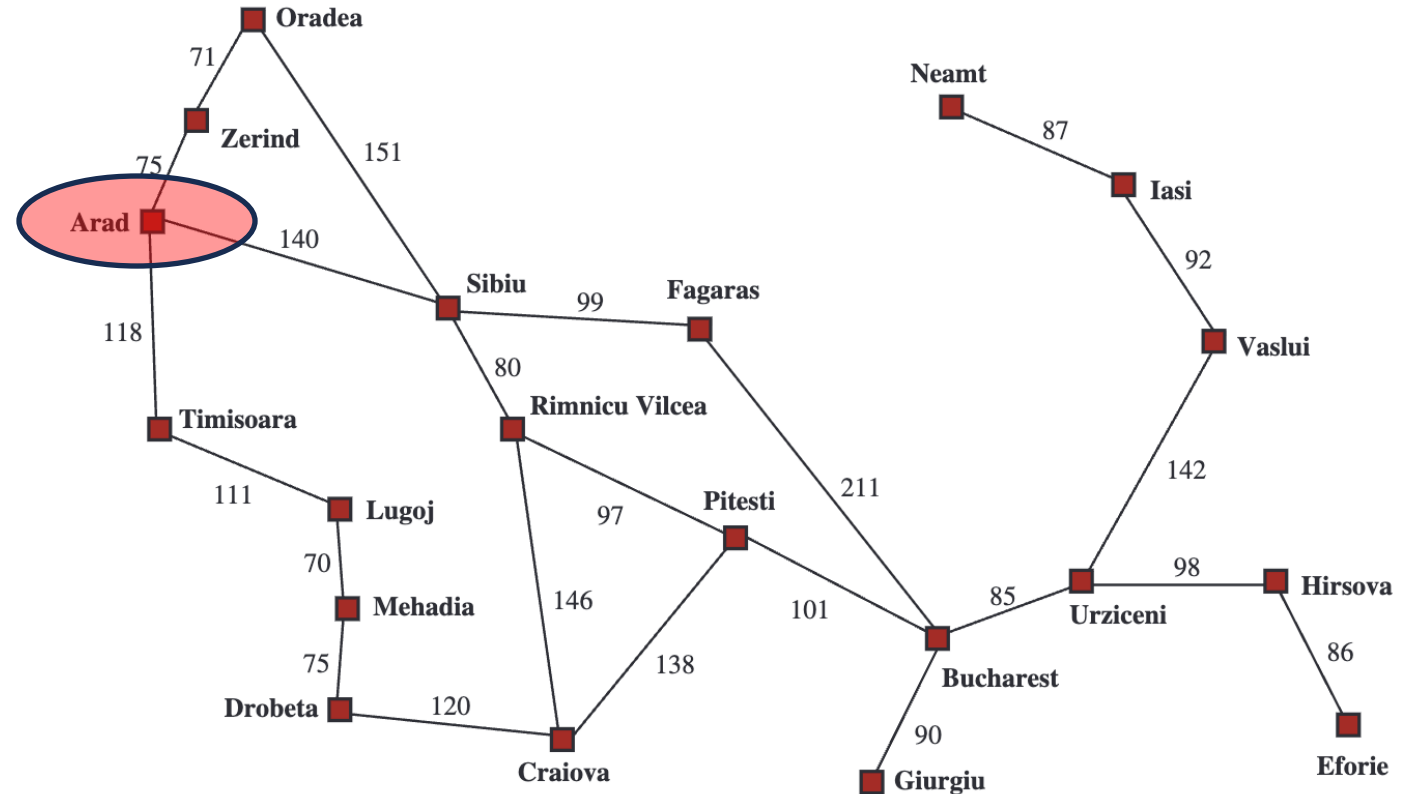


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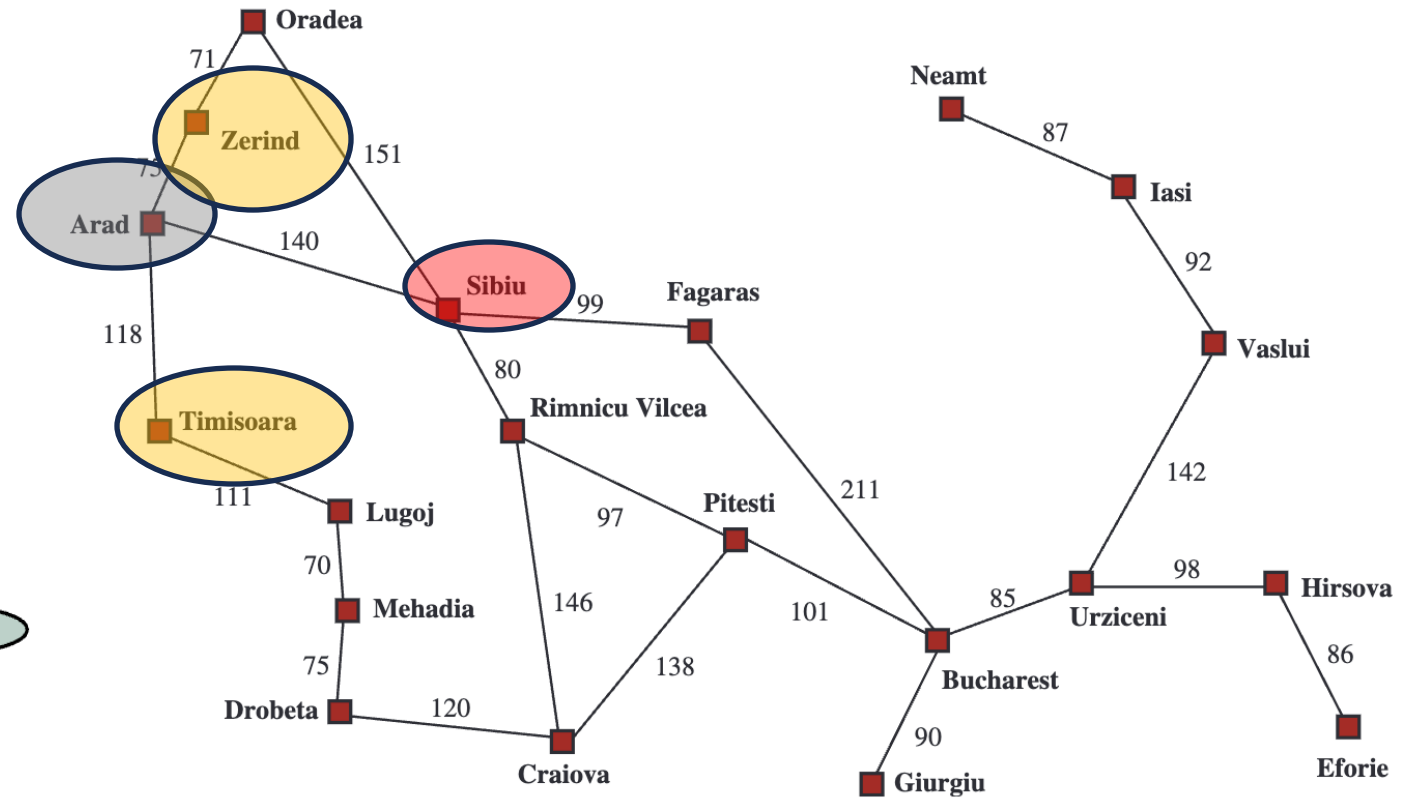
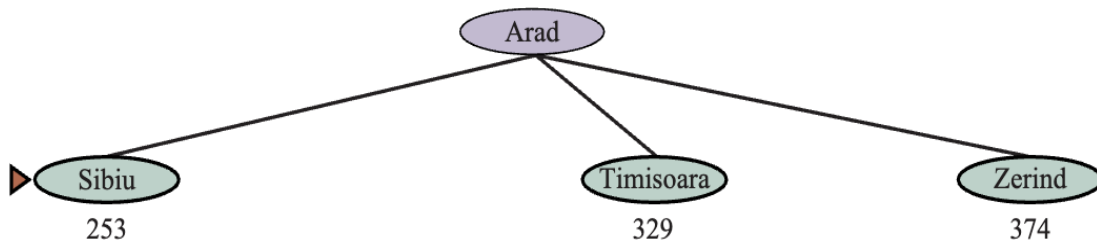


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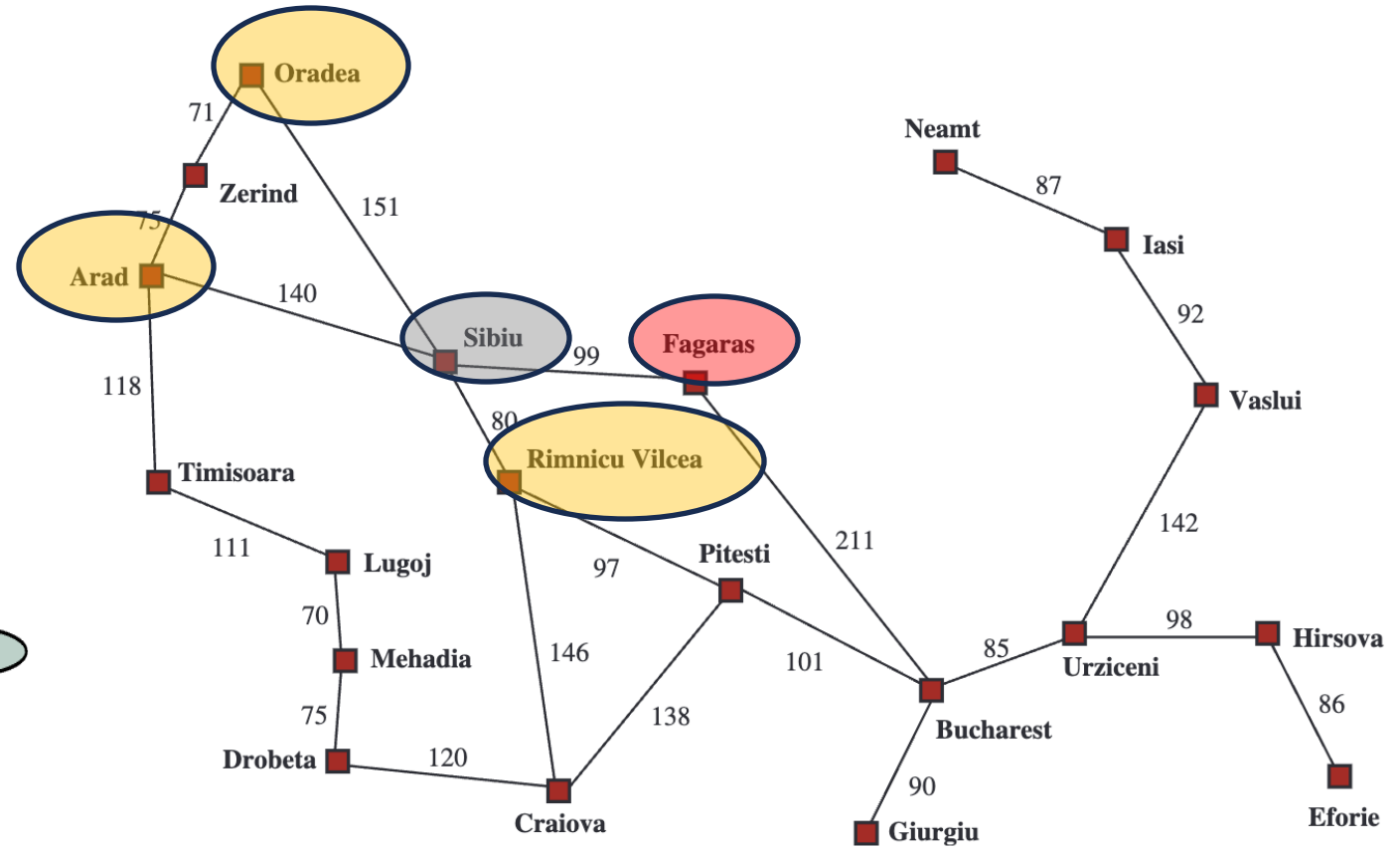
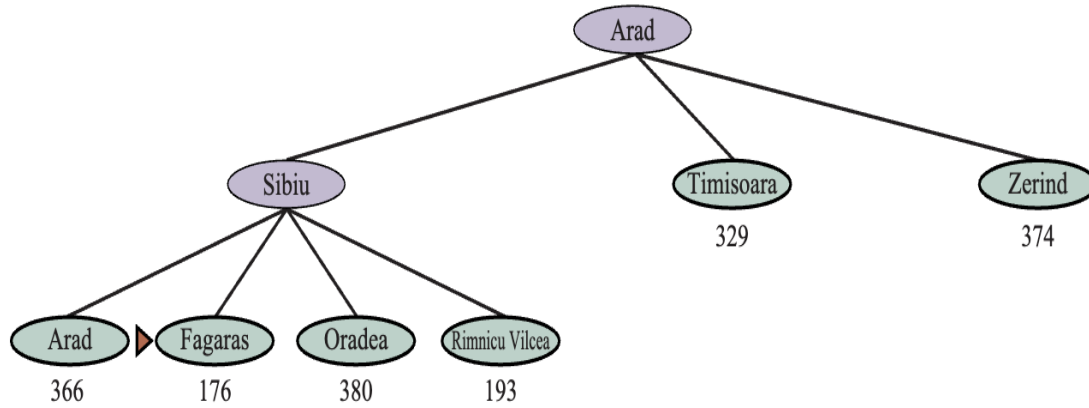
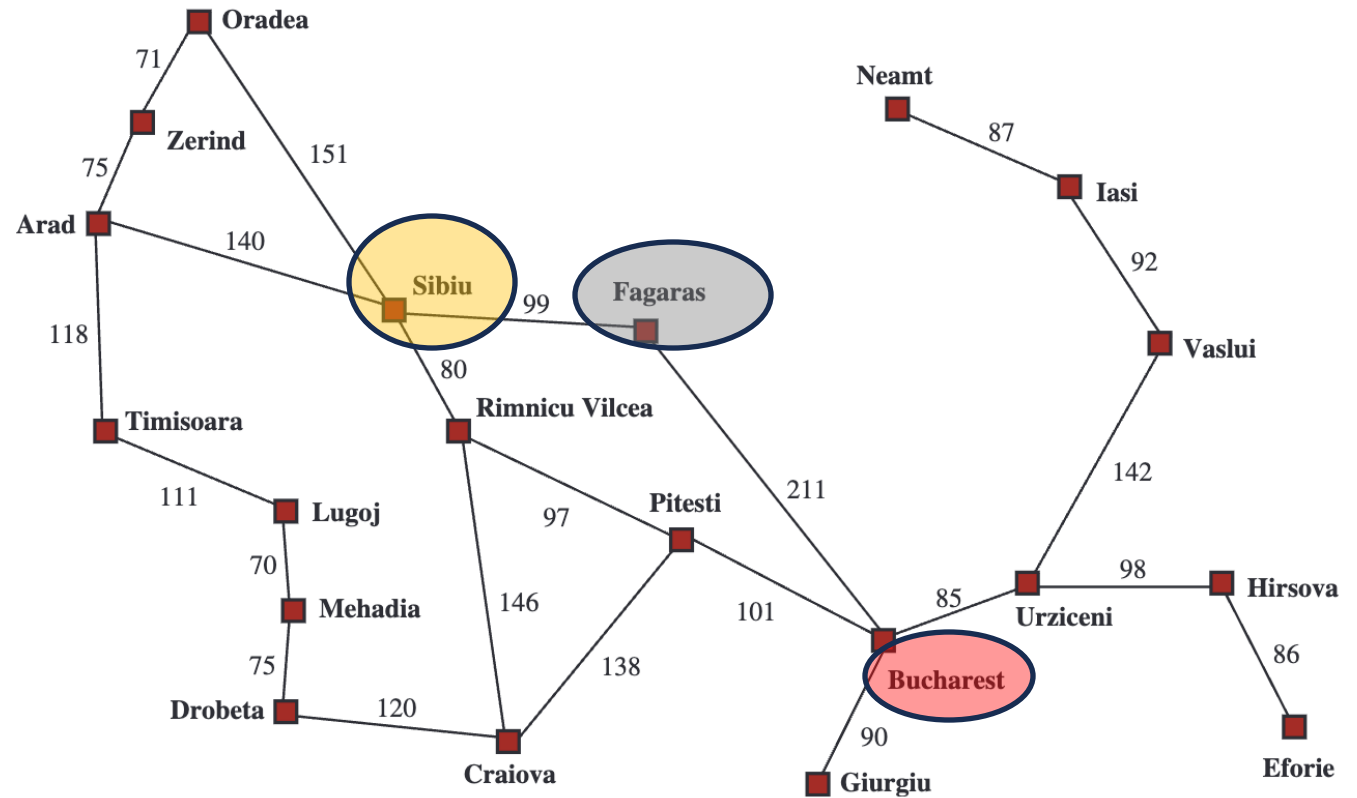
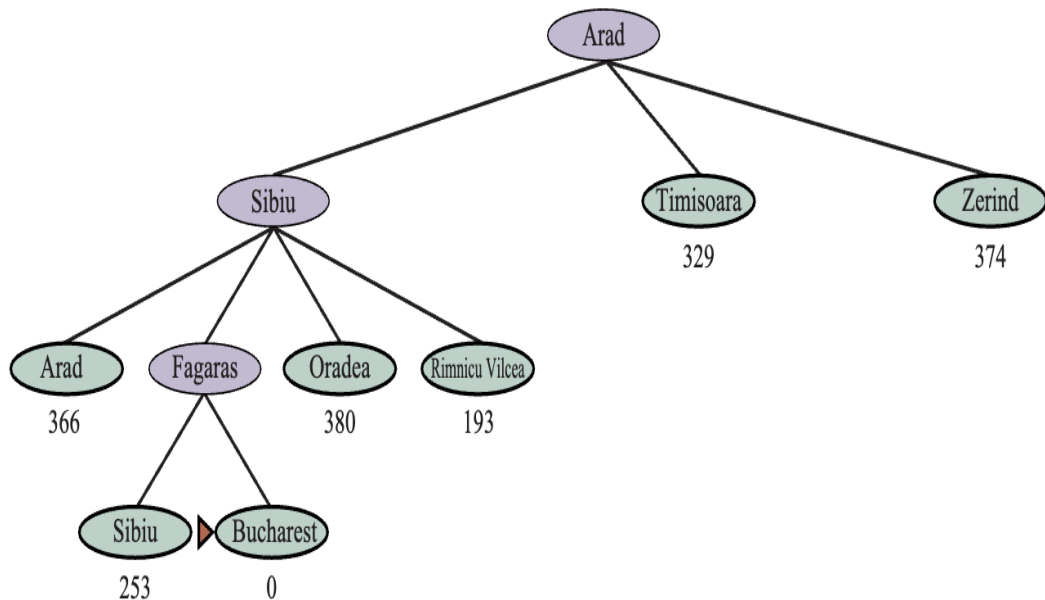


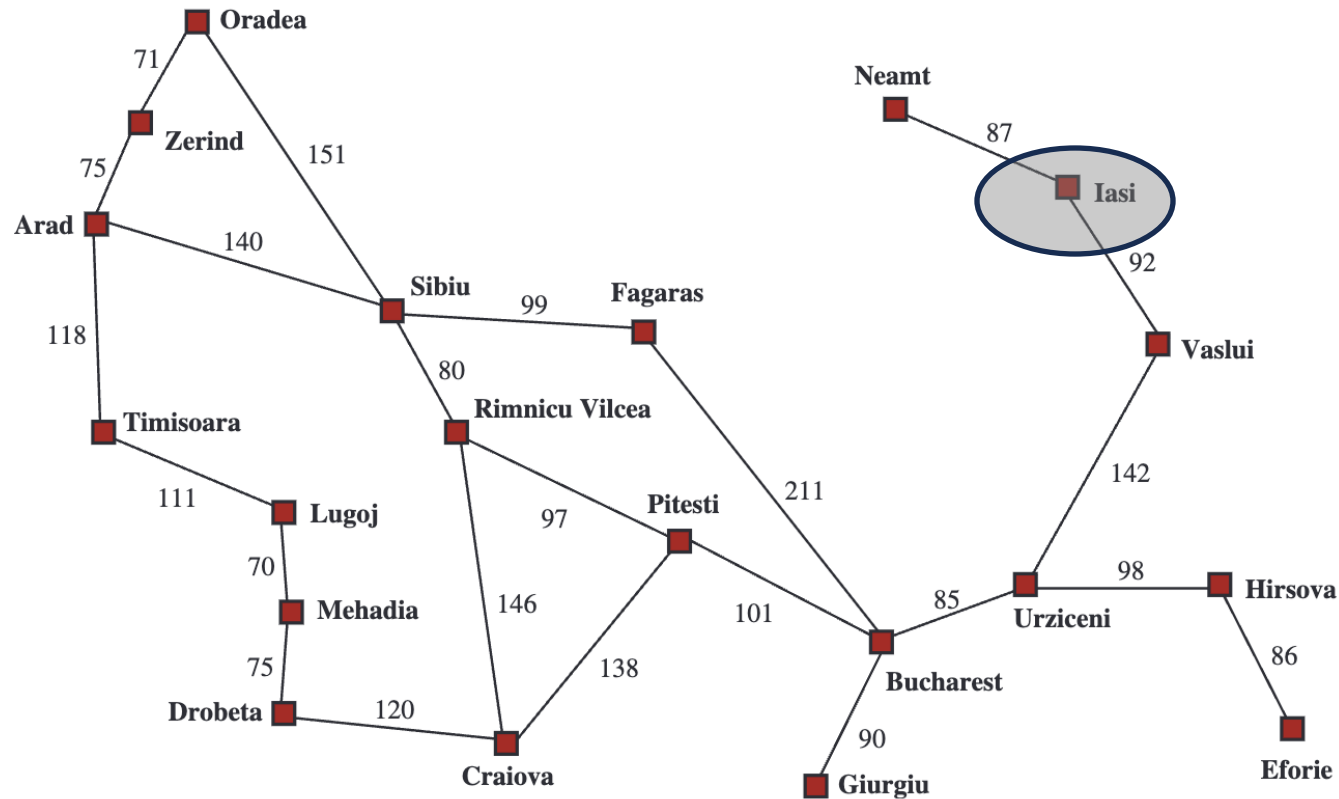
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Is greedy guaranteed to find a solution?



- No, an get stuck in loops, e.g., with Oradea as goal, Iasi $\rightarrow$ Neamt $\rightarrow$ Iasi $\rightarrow$ Neamt $\rightarrow$...



Next: A^* search algorithm

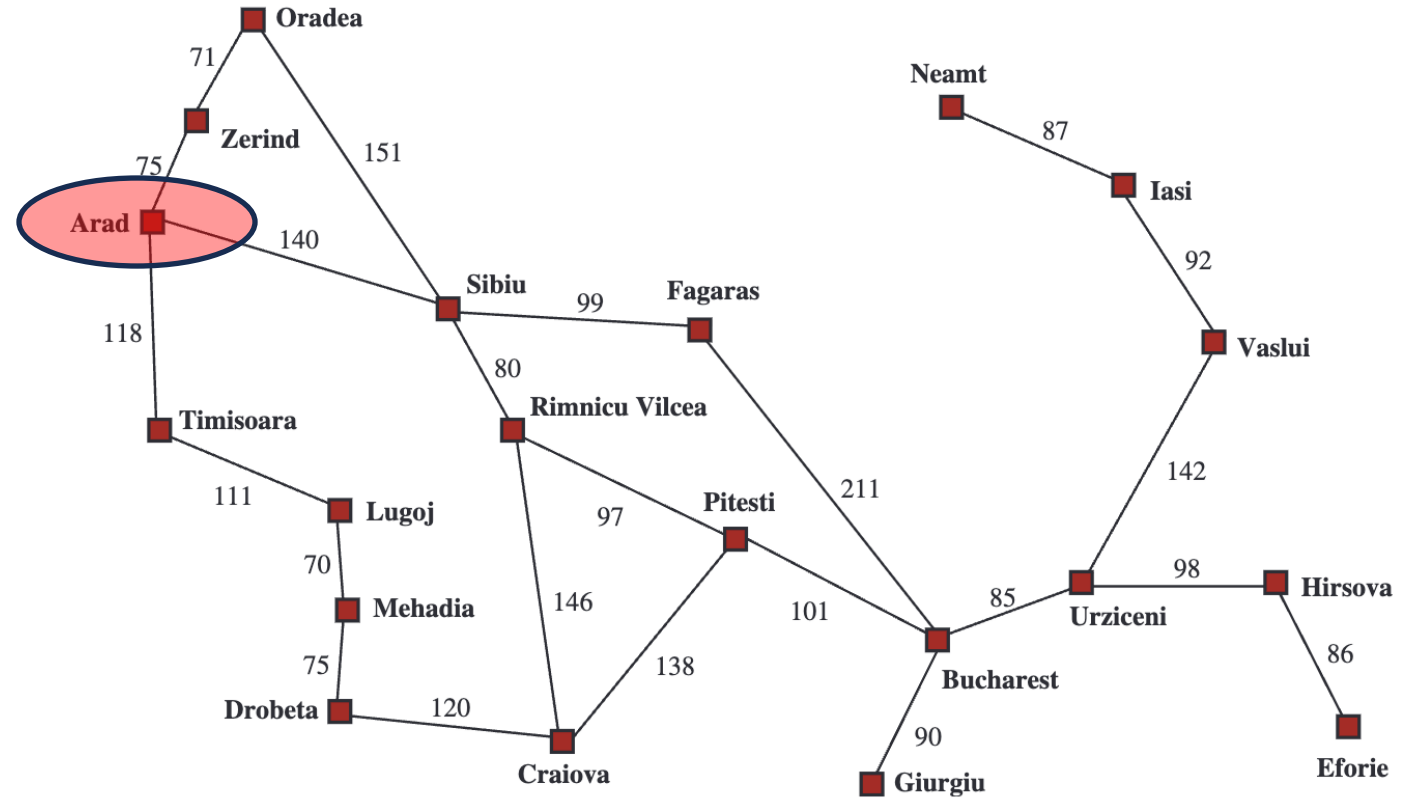
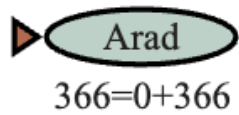
- Idea: avoid expanding paths that are already expensive
- Combine uniform cost search and greedy search: let the cost function $f(n) = g(n) + h(n)$
 - $g(n)$ is the incurred cost at step n , which ensures we account for cost so far
 - $h(n)$ is a heuristic function that estimates how much more cost we would incur to reach the goal



An illustration of A^* search

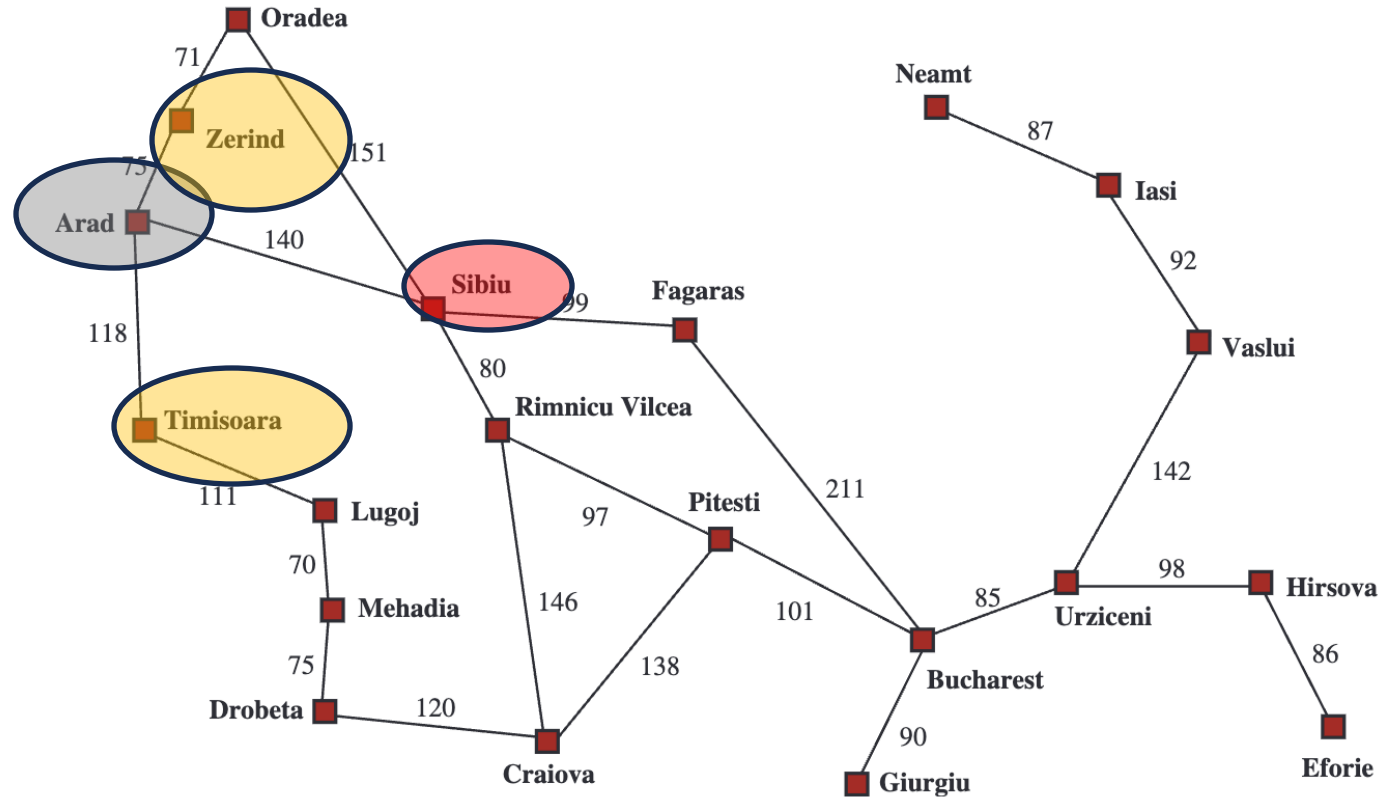
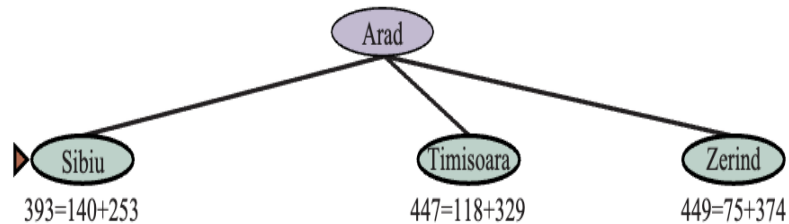
A table of heuristic functions

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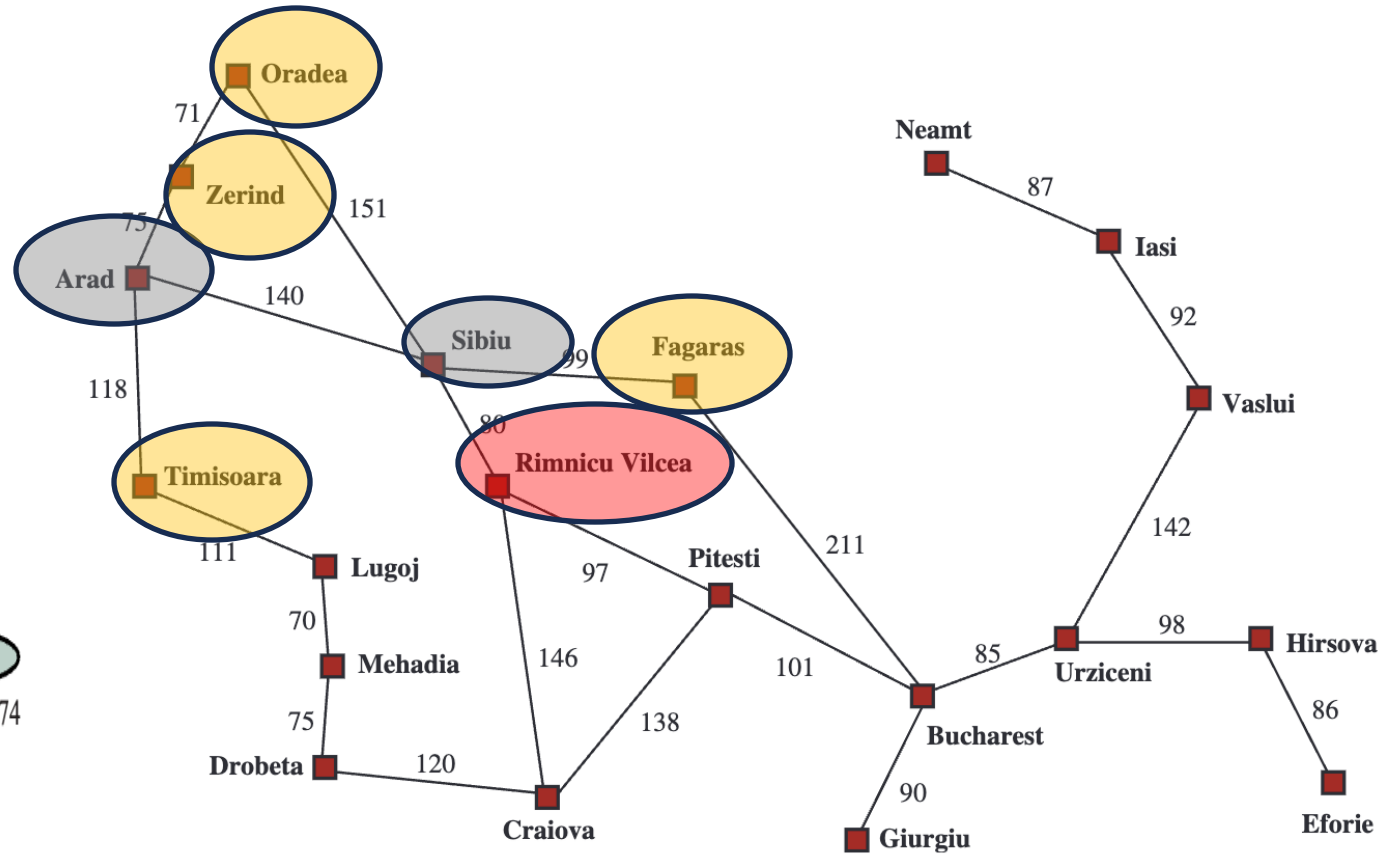
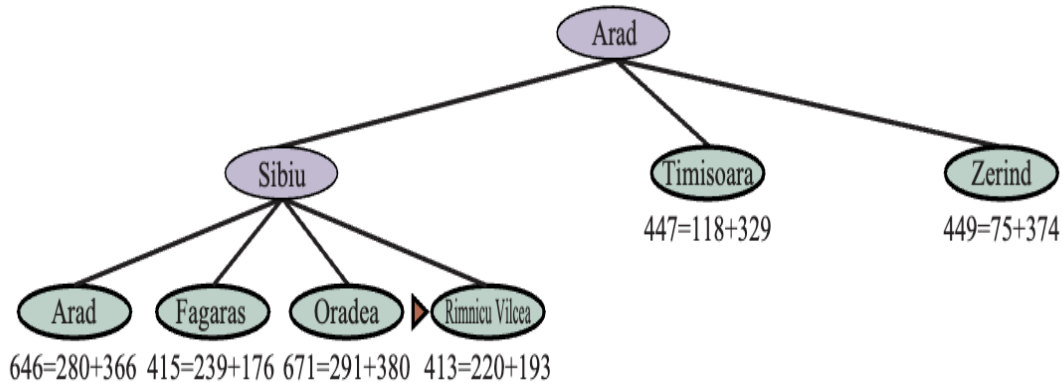
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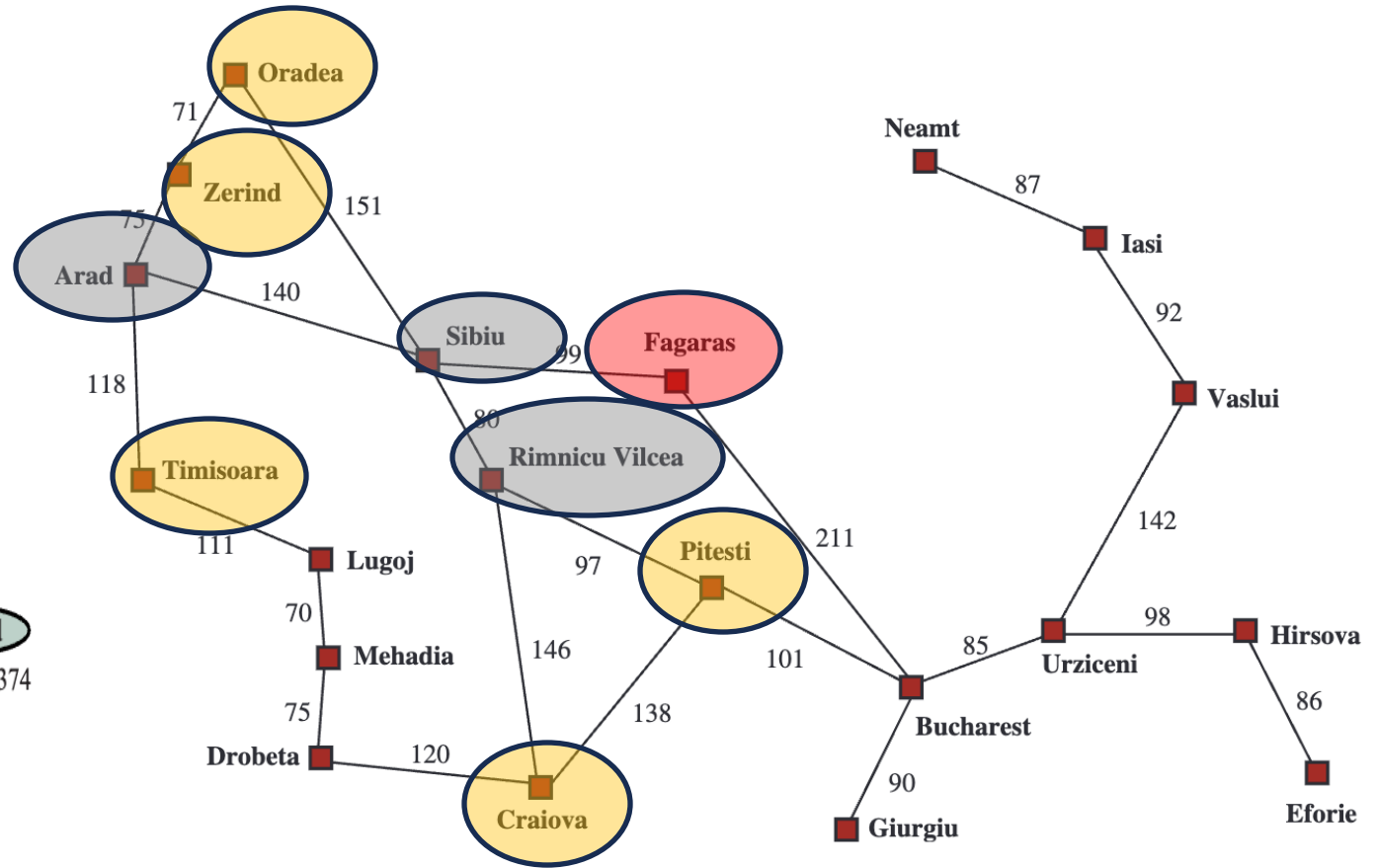
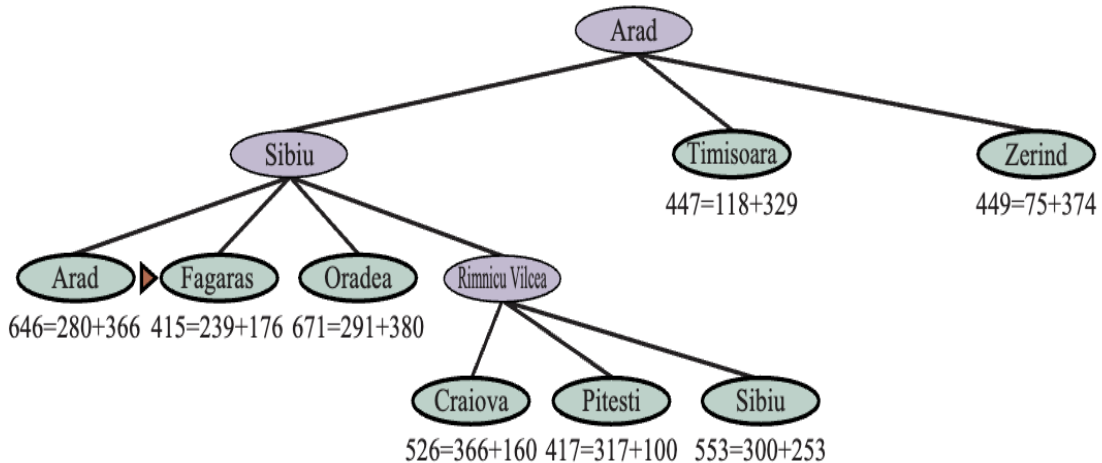
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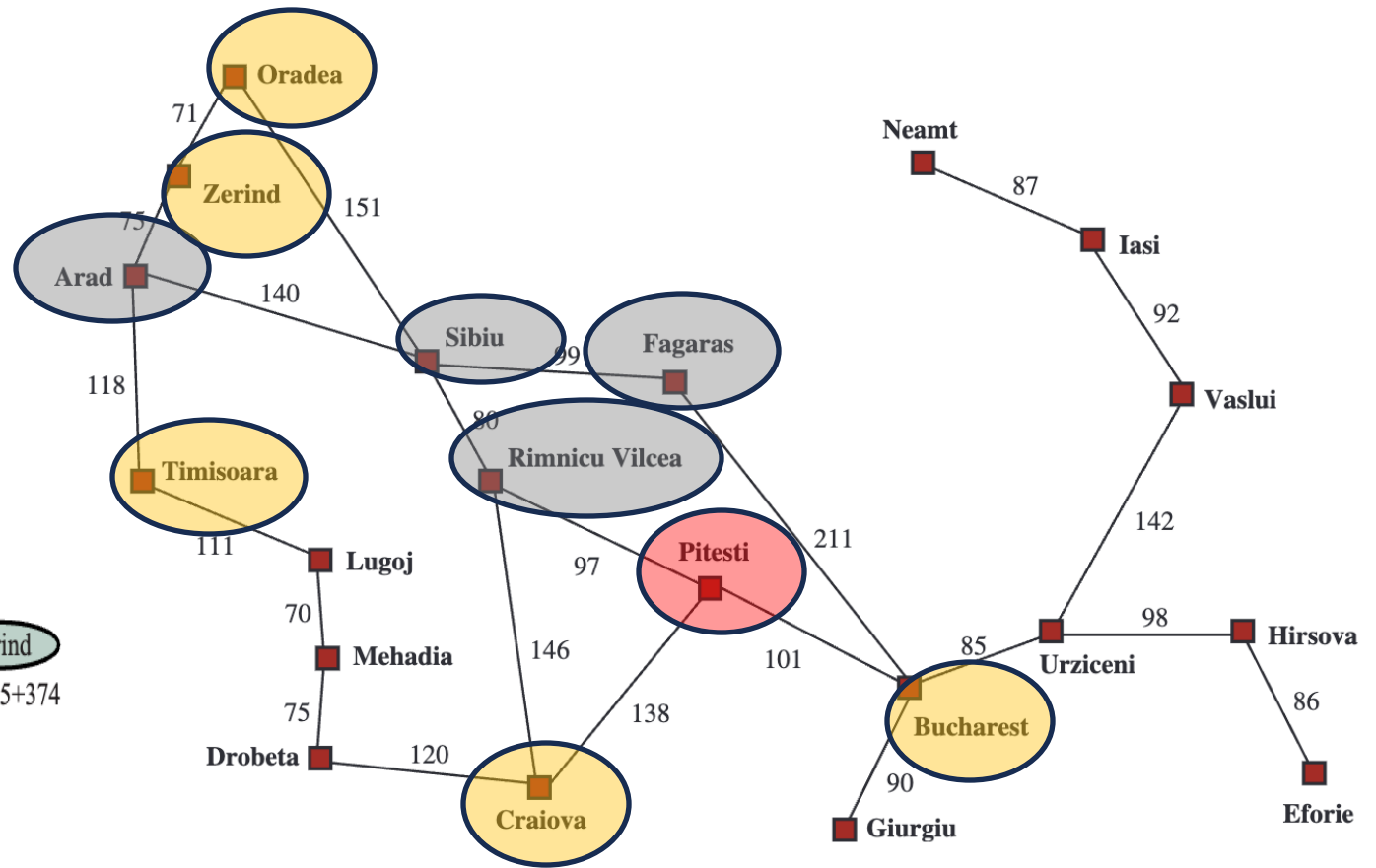
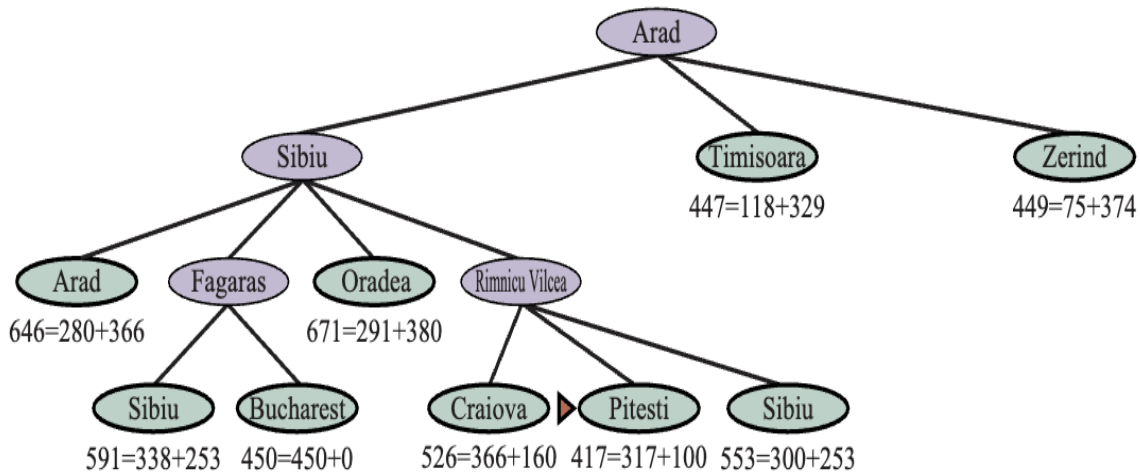
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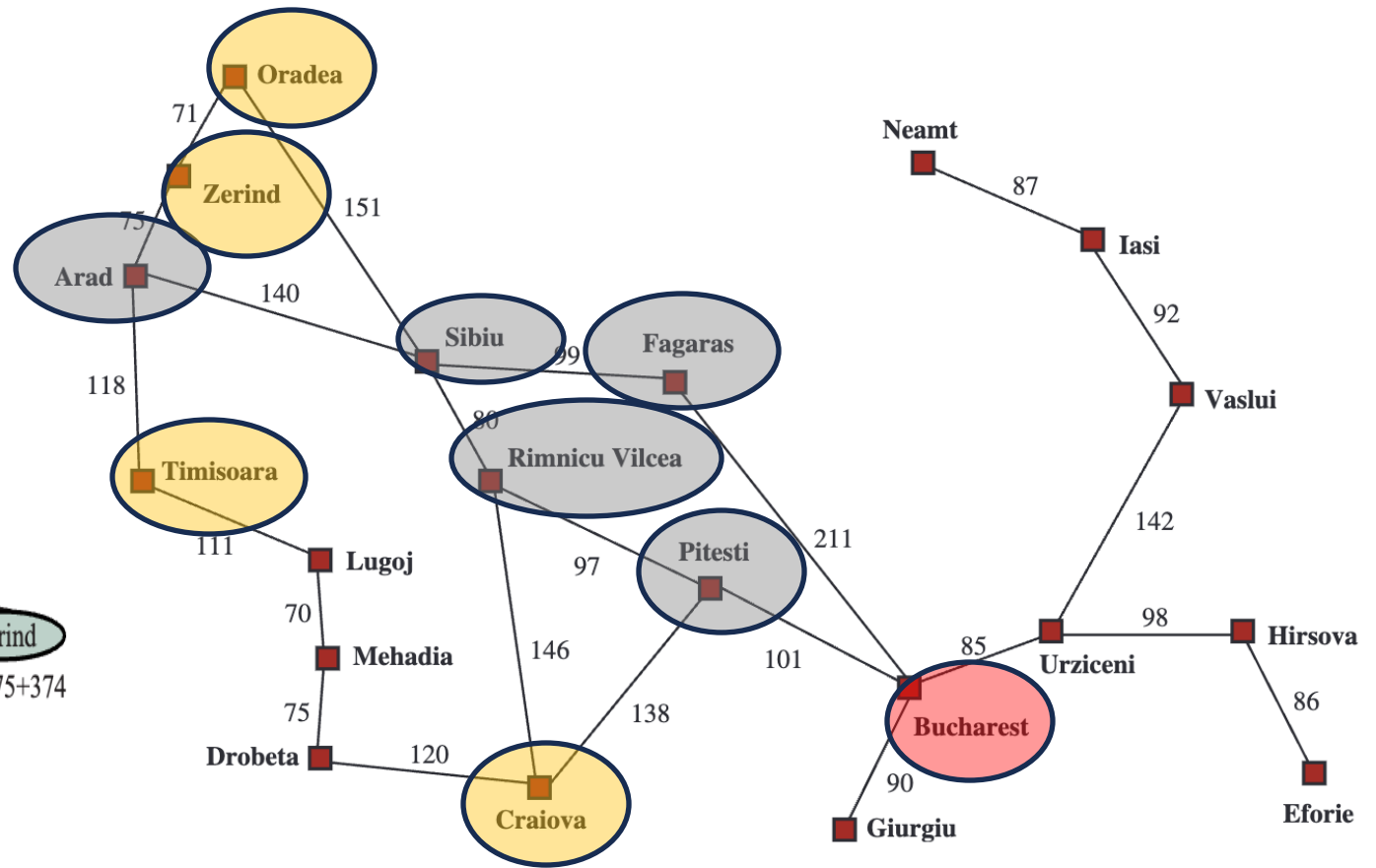
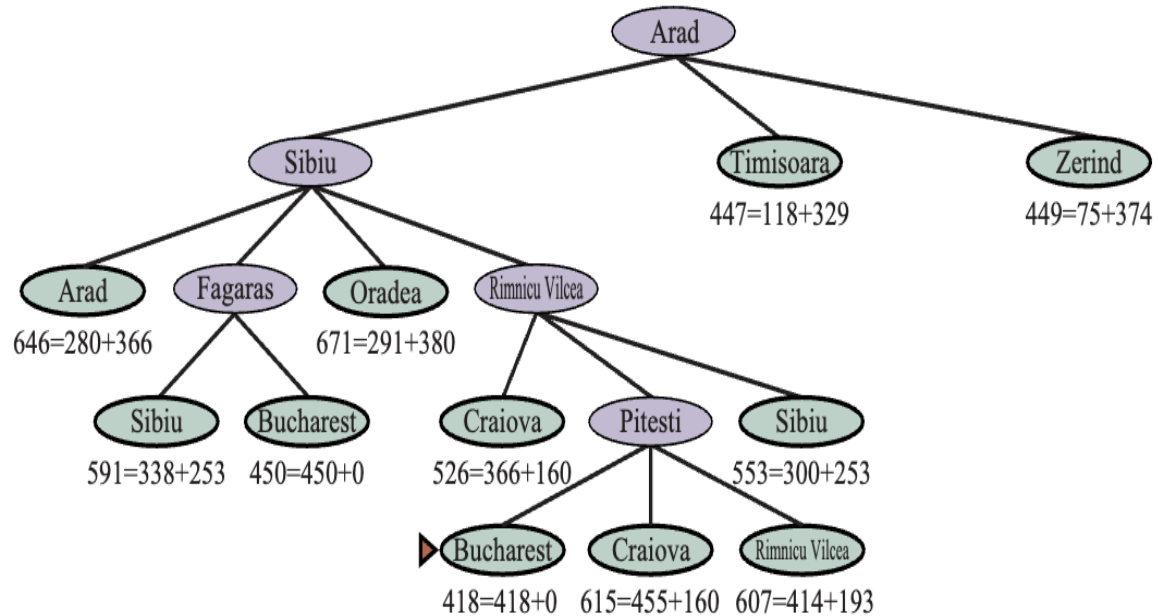
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A^* search is optimal

- **Property:** The solution found by A^* search is optimal if the heuristic $h(n)$ is admissible, meaning that it is never an overestimate of the cost from node n to a goal node
 - $\forall n, h(n) \leq h^*(n)$, where $h^*(n)$ is the true cost from n
 - Require $h(n) \geq 0$, so $h(G) = 0$ for any goal G
- **Why?** $f(n)$ never decreases along any path due to the above condition on $h(n)$, so first goal found has lowest f = lowest actual cost



Summary

- Greedy search algorithm: Can get stuck in loops, finds a solution quickly but not necessarily best
- A^* search algorithm: guaranteed to find a solution, and ensure optimality
 - Utilizes a heuristic function to help estimate the cost towards the target



Variations: Satisficing search

- Sometimes we need a solution quickly, even if not optimal
 - **Video games:** prioritize response time, such as needing to react in milliseconds
 - **Robotics:** avoid obstacles on the road spontaneously
 - **Web services:** quickly answer a query without extended wait time
- Examples
 - Deliberately overestimate to speed up search
 - Weighted A^* : inflate heuristic influence systematically
- Beam search: Memory-bounded search: Limits the size of the frontier
 - The easiest approach is to keep only the k nodes with the best f -scores, discarding any other expanded nodes



Extensions: Weighted A^* search

- $f(n) = g(n) + w \times h(n)$
 - $w = 1$: standard A^* (optimal)
 - $w > 1$: weighted A^* (faster, sub-optimal)
 - $w \rightarrow \infty$: approaches pure greedy search
- **Bounded sub-optimality:** weighted A^* with admissible h finds solution with cost $\leq w \times \text{optimal}$
 - Example: $w = 1.5$ guarantees solution within 50% of optimal



Conclusion

- Informed (heuristic) search algorithms
 - Greedy (best-first) search
 - A^* search
 - Variations: beam search, weighted A^* search
- Search algorithms are extremely useful in motion planning and path finding problems



Example roles

- Software Engineer
 - E.g., joke execution, ensure self-driving reliability, ...
 - Machine Learning Engineer
 - Building models or deep learning algorithms to learn $A \rightarrow B$ mappings
 - Machine Learning Researcher
 - Extend state-of-the-art in ML: Machine learning is still advancing rapidly
- } Applied ML Scientist



Example roles

- Data Scientist

- Examine data and provide insights
- Make presentation to team/executive

1 MB (megabyte)
1,000 MB = GB (gigabyte)
1,000,000 MB = TB (terabyte)
1,000,000,000 MB = PB (petabyte)

- Data Engineer

- Organize data
- Make sure data is saved in an easily accessible, secure and cost-effective way

- AI Product Manager

- Help decide what to build; what's feasible and valuable

